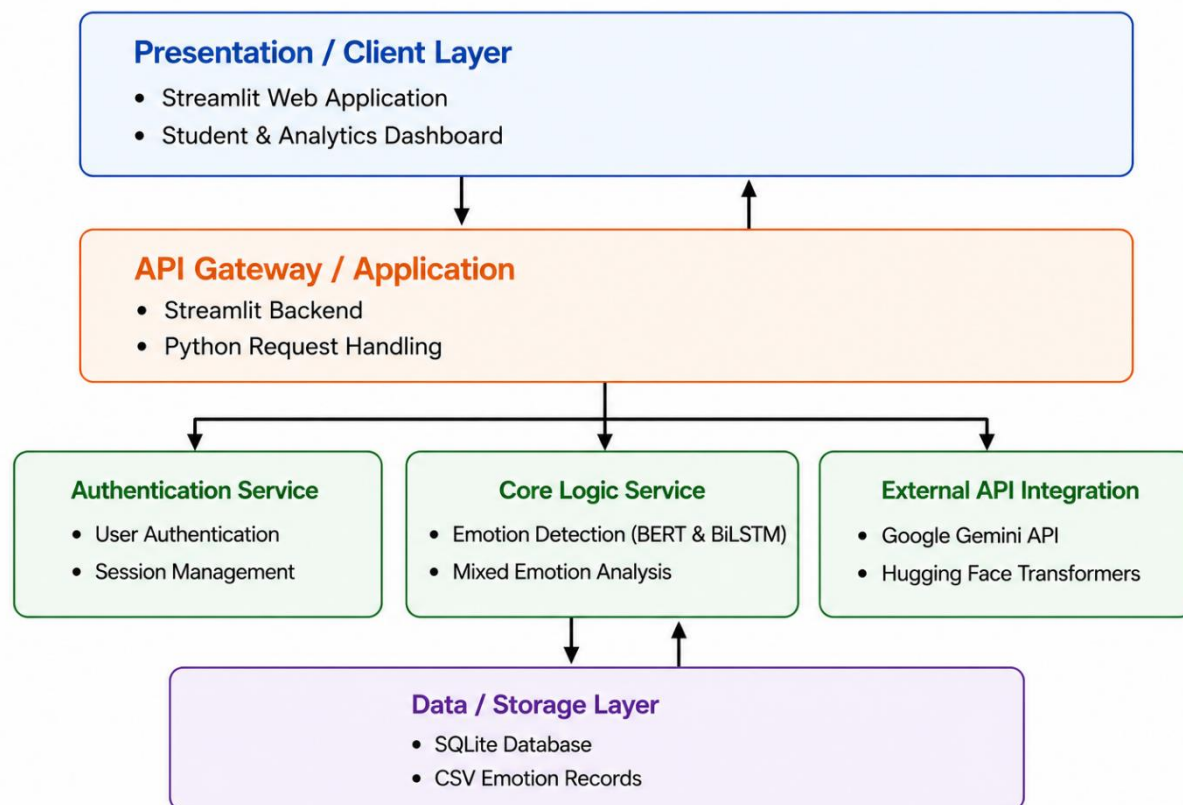


Solution Architecture

Date	15 March 2026
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	Provides the user interface for entering learning problems, displaying detected emotions, AI guidance, and analytics dashboard.	Streamlit, HTML, CSS
[API Gateway]	Handles requests from the user interface, manages sessions, validates inputs, and forwards data to the emotion detection services.	Python, Streamlit Backend
[Microservices]	Performs text preprocessing, emotion prediction using BERT and BiLSTM models, mixed emotion analysis, confidence calculation, and Gemini API integration.	Python, PyTorch, Transformers, Gemini API]
[Database]	Stores student emotion records, prediction history, datasets, trained models, and analytics information.	SQLite, CSV Files